Low Voltage Grounded Auxiliary Power Unit

AUTO1931 — Scope Report

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1 Overview

Formula SAE Electric (FSAE-E) vehicles have two main potential levels: one is the tractive system, which is up to 600 VDC, and the other is the low-voltage grounded (LVG) system, which is up to 60 VDC. RMIT Motorsports (RMIT-M) uses a 600 V tractive system and a 13.2 V LVG system. The tractive system is supplied by the tractive battery, which contains 143 lithium cells. Similarly, the LVG system is supplied by a lithium battery. However, it is a commercial unit certified by the FAA for aircraft!?

The low-voltage grounded auxiliary power unit (LVG-APU) is a proposed design to feed the LVG system from the tractive system while the tractive battery is installed in the vehicle. This APU would also have to be able to supply power while the tractive battery is disconnected from the vehicle, so it is also proposed that the LVG battery is scoped within the component. Hence the name “auxiliary power unit,” as it provides the power required (LVG battery) during setup to close the contactor (similar to cranking amps), which is required to power the isolated (split grounds) DC-to-DC converter.

Later, the LVG battery buffers the line voltage during heavy loading, reducing the converter’s required power output range. This also allows for the implementation of isolated external power input for the LVG battery, which is proposed to be an IP-rated (ingress protection) USB-C input to remove the dependency of another charging apparatus within the supporting equipment. This allows for a design with high self-reliance. At worst, it could be a USB-C laptop charger!

> **Supporting:** The impedance (L, R) between the LVG battery and the DC-to-DC converter is important due to the coupling (mutual influence) between them when supplying the line voltage across a wide range of power draws and rates of change of that draw. Hence the inclusion within the APU system boundary.

Contents

2 System Specification

Here, you are to specify what your system is, the targets of this system and identify the major system design constraints. This section is to serve as the effective Problem Statement for your project. As such, it must be clear and obvious what the objectives of your system are.

3 Literature and Design Review

With your problem developed, you may now present, analyse and interpret external literature related to your project. It is anticipated that all presented literature will assist in informing the overall and specific details of your design work — if the literature does not inform your work, do not include that literature.

3.1 Example of Literature

Your literature and design review may include:

- Design reports from former RMIT FSAE teams and other Formula SAE teams
- Technical reports
- Key information from relevant textbooks
- High quality journal or technical papers from reputable sources
  - Do not review reports from low quality sources (e.g. pay-to-publish academic journals).
- Reviews of existing eminent — or otherwise — designs relevant to your project
  - This can include social media posts and team or personal photography
- Review of relevant production techniques, including materials, manufacturing process and off-the-shelf components.

3.2 Interpretation

Do not simply present the literature. You are to rigorously review the literature in the context of your problem. It is incumbent on you — as the design engineer — to interpret the literature and communicate the influence of that literature on your project. This may include, and is definitively not limited to, the critical analysis of designs from other Formula vehicles, evaluation of the Overall Vehicle Specification on your system or concise summaries of engineering science and theory pertaining to your system.

4 Conceptual Design

At this stage of the report, it should be clear what the requirements of your system are (from System Specification section) and what the engineering context of your design problem is (from the Literature and Design Review).

Now you are to present and discuss the overall concept of your system. You must indicate how the features of the concept intend to meet the targets of the system, in a general manner. You may make use of appendices to communicate information or present figures which do not fit into the page limitations of the report.

5 Project Management

Your Project Management plan should consist of a definition of project deliverables, an analysis of required resources (which may include materials, knowledge and data from RMIT's FSAE teams) and a timeline (preferably Gantt Chart) for the project. Please ensure that the timeline is effectively in visually communicating when you intend for events to occur.

The focus of this section should be regarding the operation and completion of the design cycle.

6 Conclusion

Summarise the outcomes of the report. Nominate any areas where particular progress or novel outcomes have been achieved.